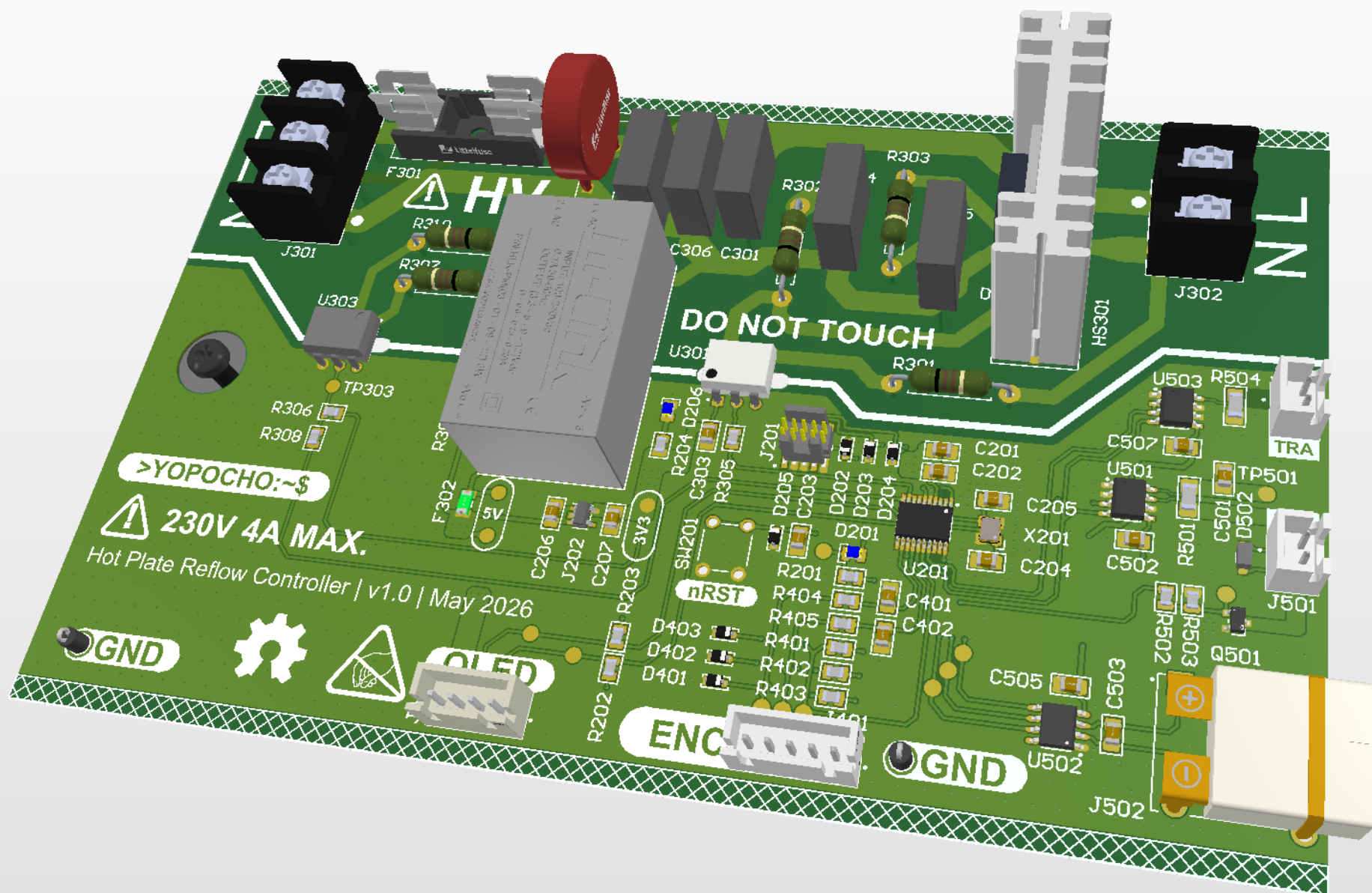
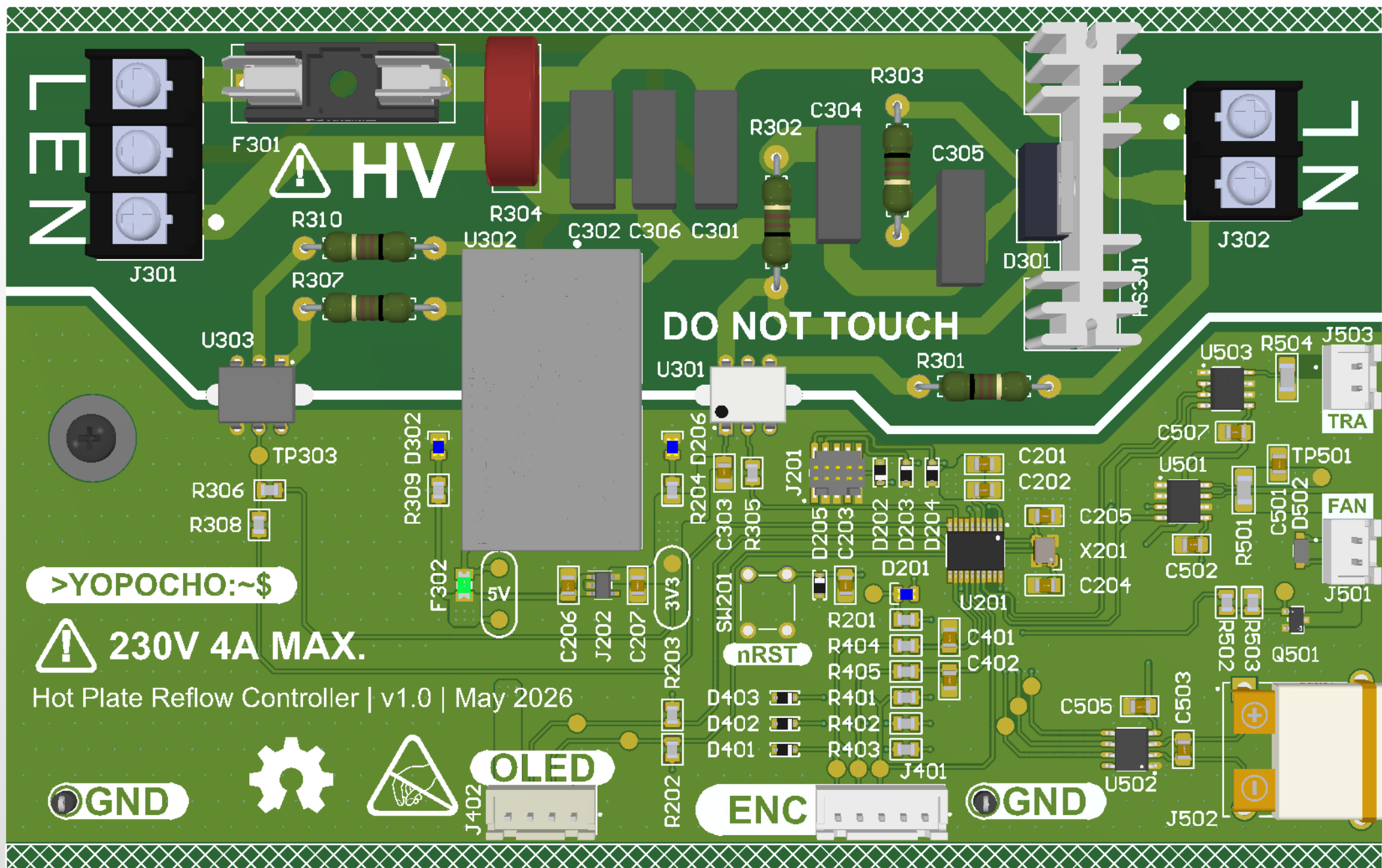




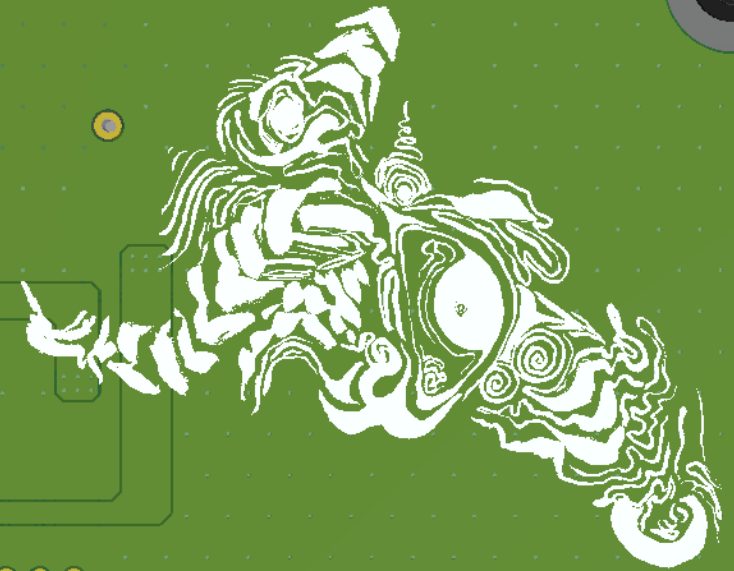


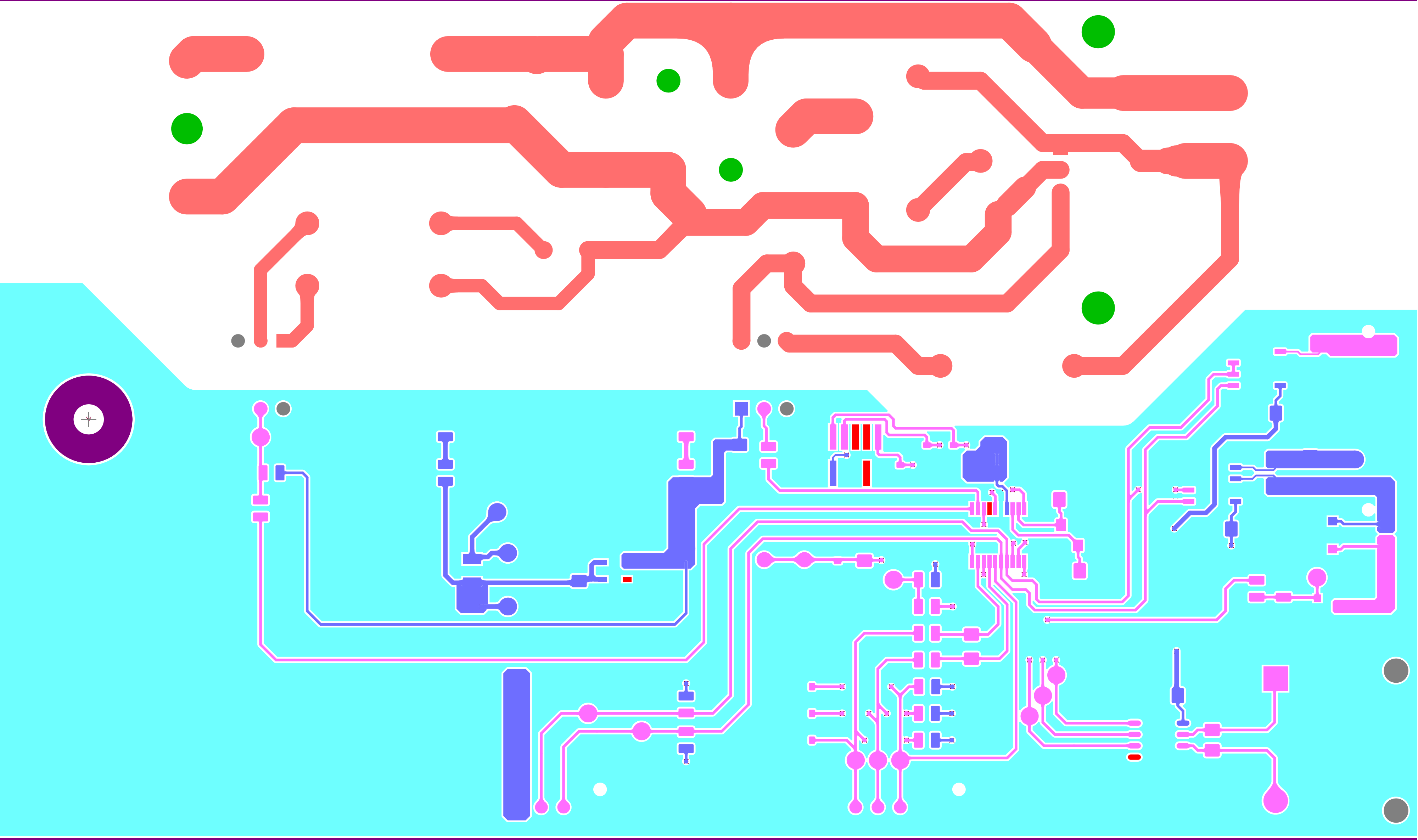
DO NOT TOUCH

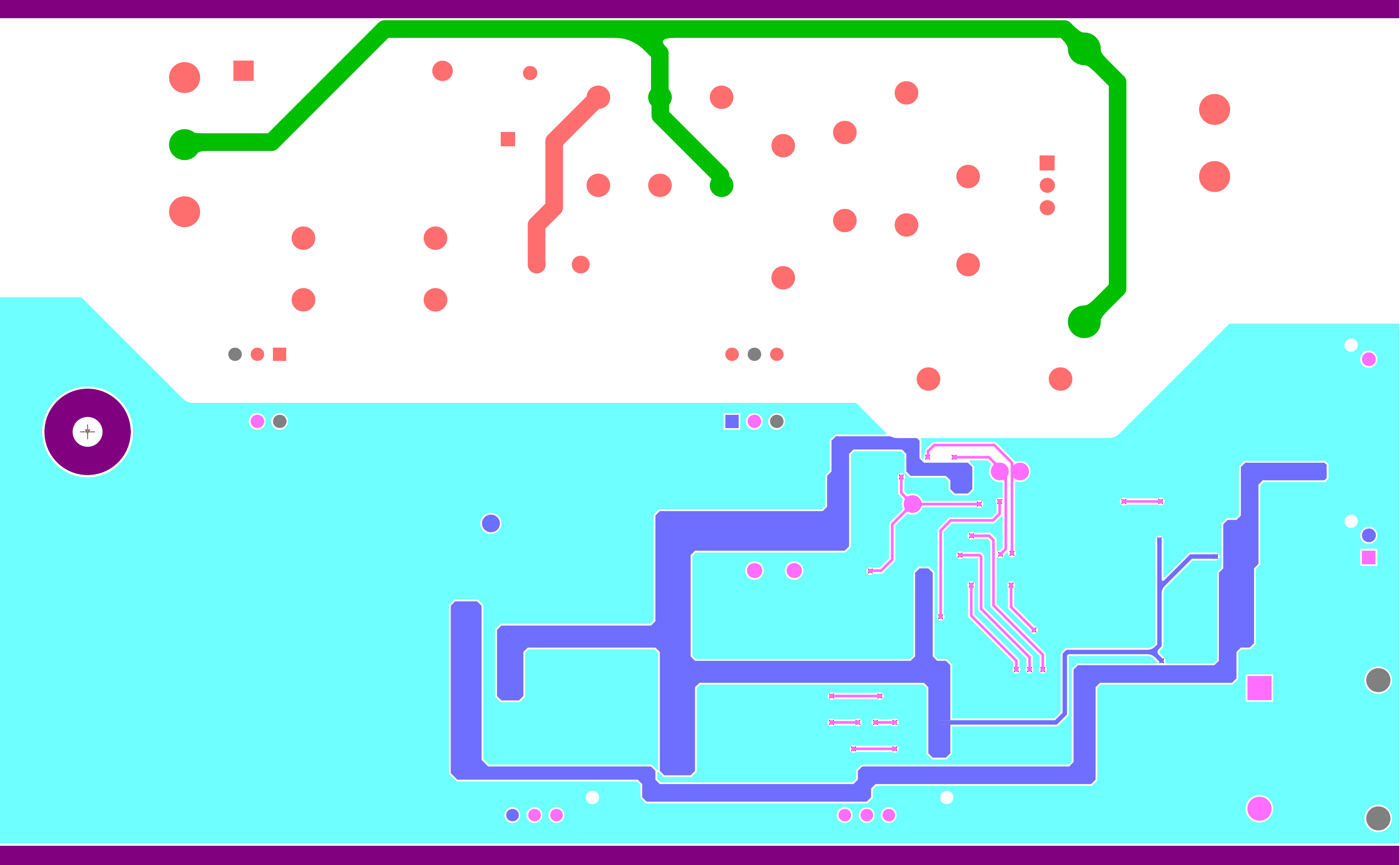
HV



GITHUB





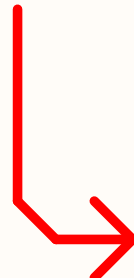


Hot Plate Reflow Controller

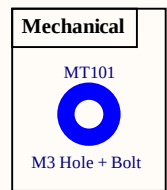
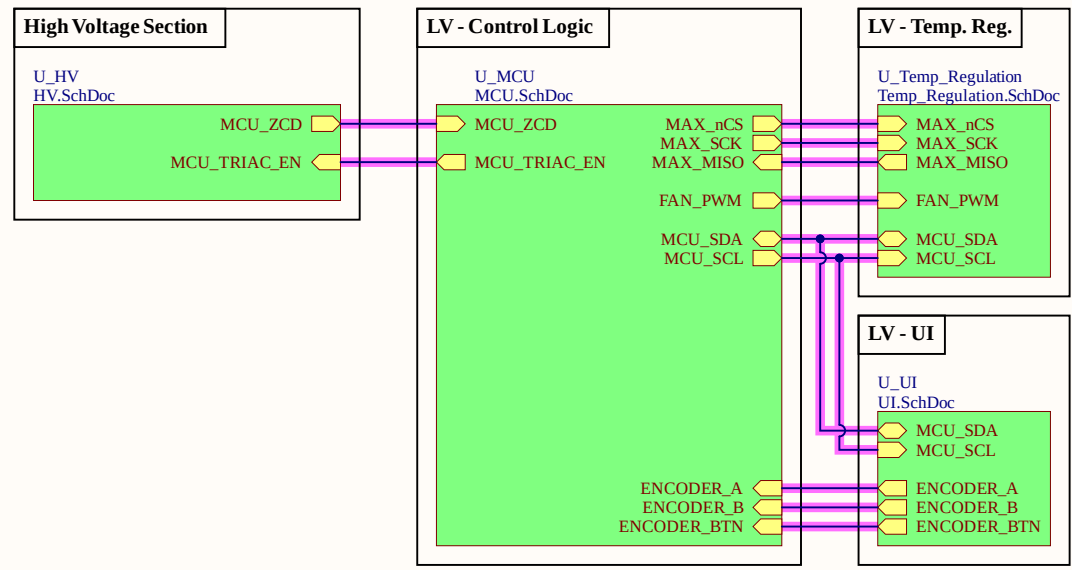
Designed to function in conjunction with electric flat-top griddle

Features:

- Closed-loop control utilizing thermocouple
- Simple addition to existing griddle
- Selectable temperature or follow reflow curve
- Compact
- Open Source!



Example



All resistors $\pm 1\%$ unless stated otherwise
All capacitors $\pm 10\text{-}20\%$ unless stated otherwise



open source
hardware



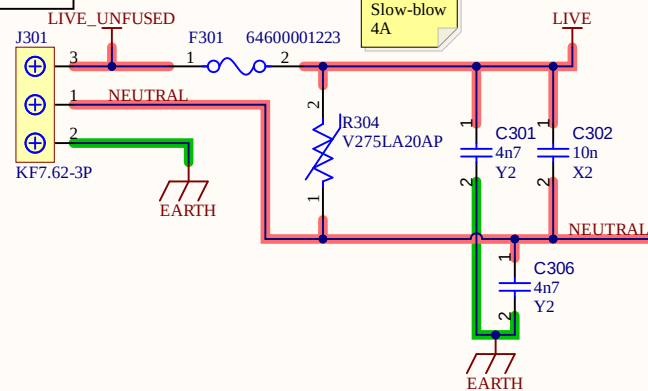
ATTENTION
HOT SURFACE

Links

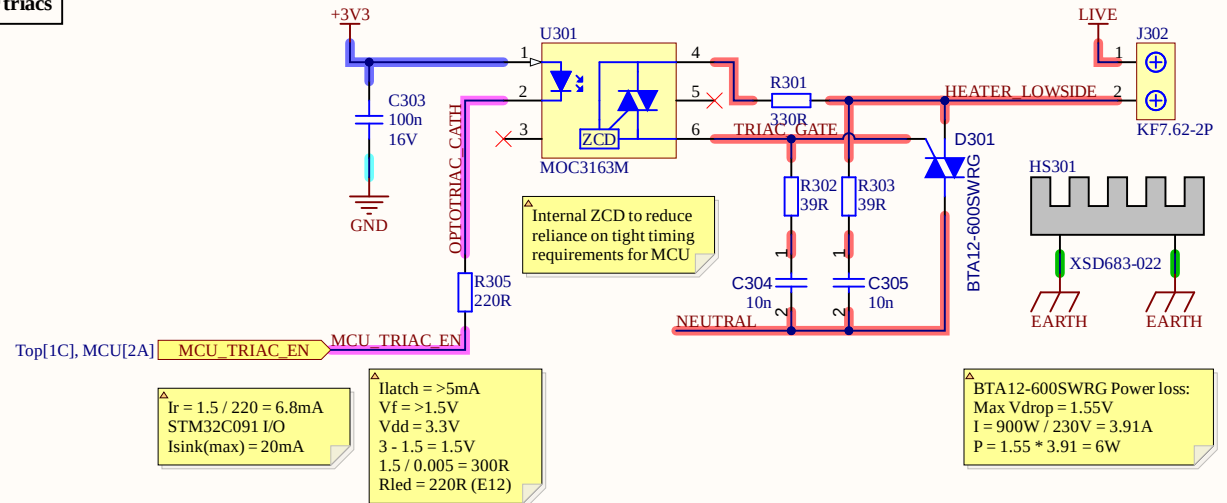
<https://github.com/yopocho/HotPlateReflowController>

Title Hot Plate Reflow Controller		
Size A4	Number Top.SchDoc	Revision v1.0
Date: 5/31/2026	Sheet 1 of 5	
File: C:\Users\...\Top.SchDoc	Drawn By:	

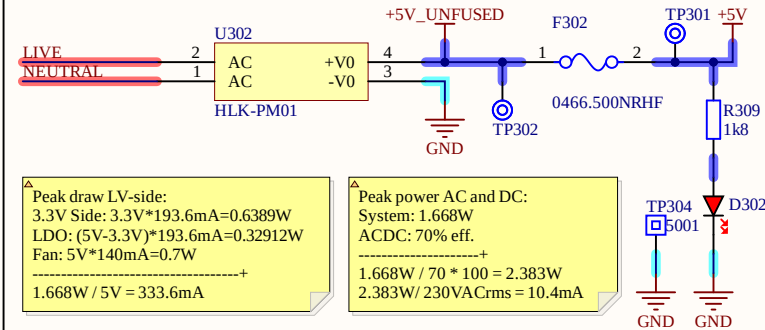
Mains in



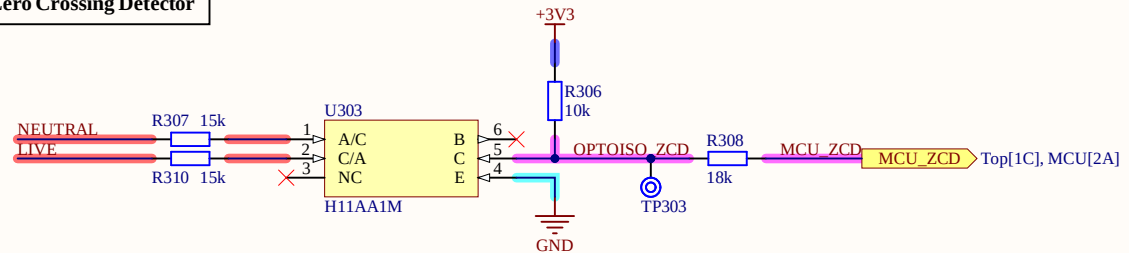
Heater triacs



5V PSU



Zero Crossing Detector



Links

https://www.laskakit.cz/user/related_files/hlk-pm01-hi-link.pdf
<https://www.onsemi.com/download/application-notes/pdf/an-3003cn.pdf>
https://www.st.com/resource/en/application_note/an437-rc-snubber-circuit-design-for-triacs-stmicroelectronics.pdf
<https://datasheet.octopart.com/BTA12-600SWRG-STMicroelectronics-datasheet-8492128.pdf?src=supplier=Factory+Futures>
<https://datasheet.octopart.com/H11AA1M-ON-Semiconductor-datasheet-8399095.pdf?src=supplier=Upverter>

Title

Hot Plate Reflow Controller

Size

A4

Number

HV.SchDoc

Revision

v1.0

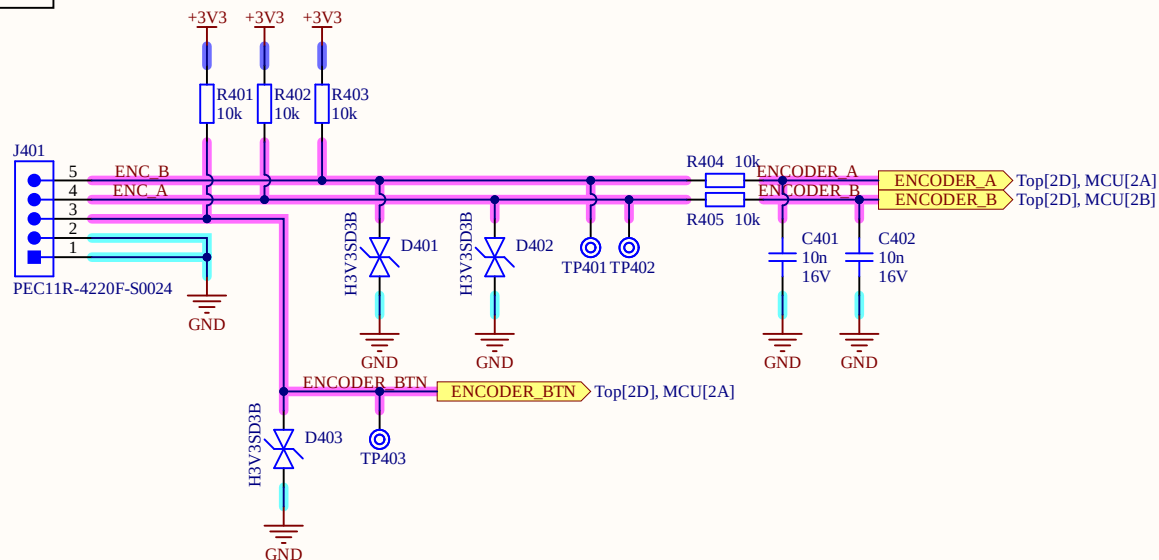
Date: 5/31/2026

Sheet 3of 5

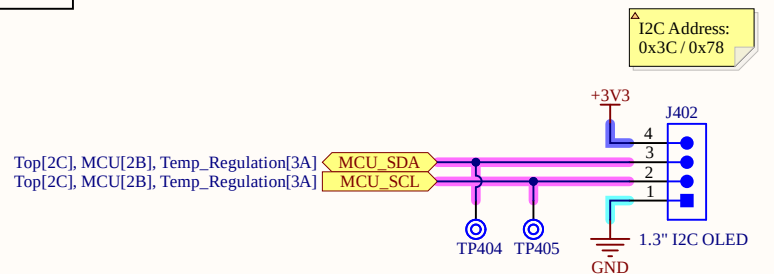
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Drawn By:

Encoder



OLED

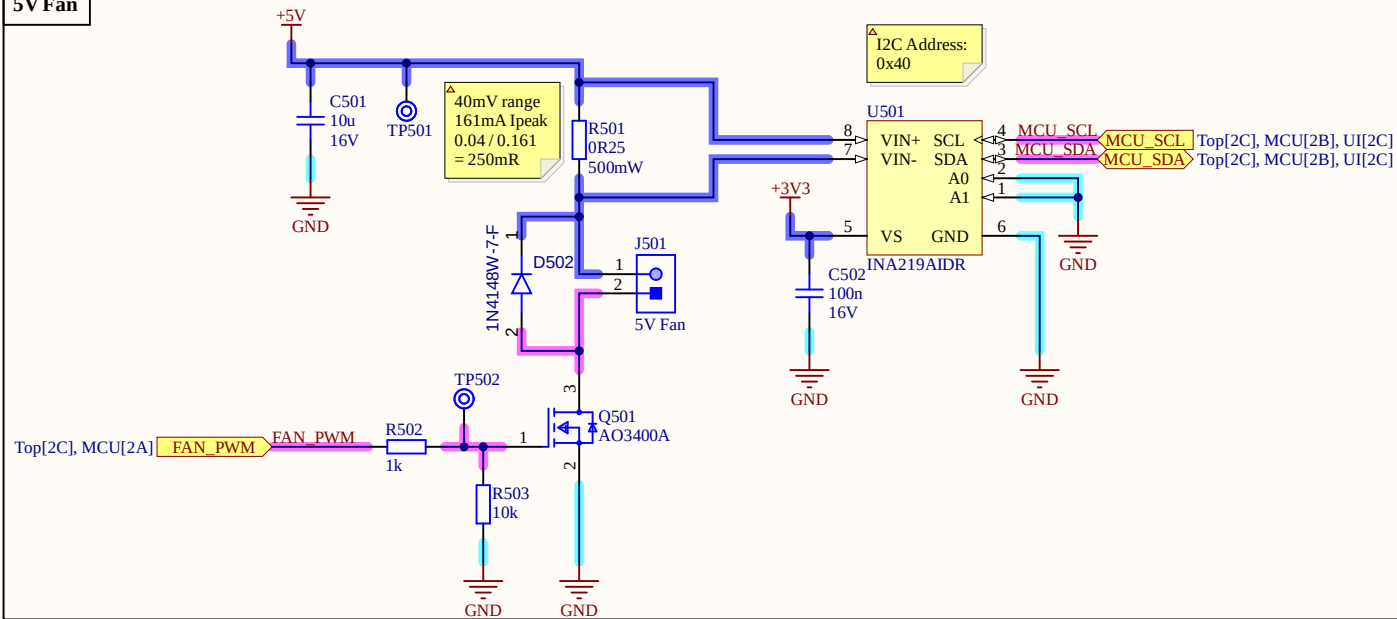


Links

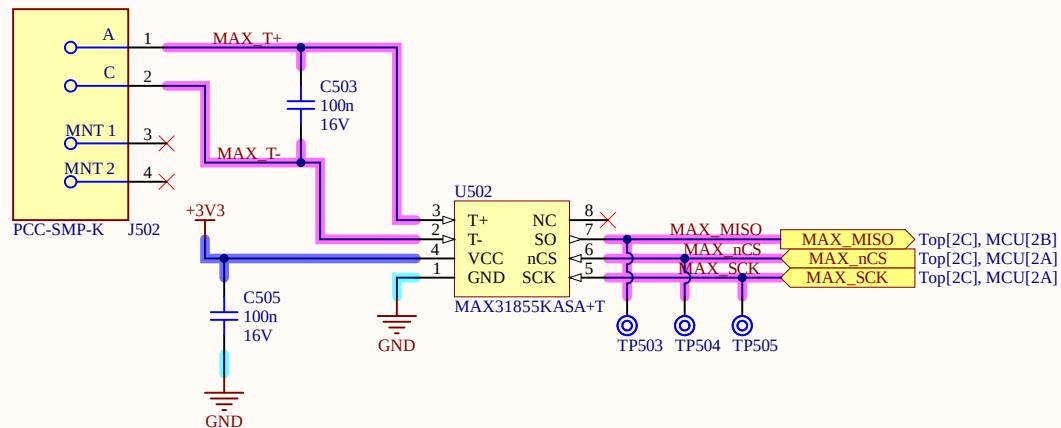
https://www.tinytronics.nl/product_files/001709_Data_Sheet_of_1.3_inch_OLED_Display.pdf
<https://mouser.com/datasheet/3/40/1/PEC11R.pdf>

Title		
Hot Plate Reflow Controller		
Size	Number	Revision
A4	UI.SchDoc	v1.0
Date:	5/31/2026	Sheet 4 of 5
File:	C:\Users\...\UI.SchDoc	Drawn By:

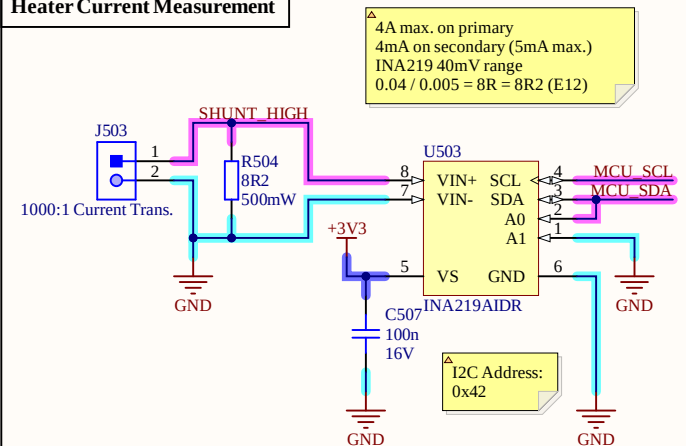
5V Fan



Thermocouple



Heater Current Measurement



Links

<https://www.analog.com/media/en/technical-documentation/data-sheets/MAX31855.pdf>
<https://www.farnell.com/datasheets/2339550.pdf>
<https://www.ti.com/lit/gpn/ina219>

Title Hot Plate Reflow Controller

Size A4	Number Temp_Regulation	Revision v1.0
Date:	5/31/2026	Sheet 5of 5
File:	C:\Users\...\Temp_Regulation.SchDoc	Drawn By: